Contextual predictability in negation is not special: General facilitation but no structural effects

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Prediction has been a prominent notion in language comprehension research for more than two decades. A robust finding is that the contextual predictability of a word reliably modulates multiple dependent measures reflecting different levels of language processing, from syntax-semantics to phonology.[1–4] When it comes to negative sentences, however, previous research suggests a more circumscribed role: predictability does not seem to influence the well-known crossover interaction between sentence polarity and truth value that is typically reported in the negation literature[5] (e.g., faster verification for "A robin is a bird" than for "A robin is a tree" but the reverse for their negative counterparts). Another (indirect) predictability manipulation, boosting an object's salience in a visual scene, only reduces the cost of negation under specific pragmatic circumstances, such as when the question under discussion is polar.[6]

Understanding the role of predictability in negation processing therefore presents a crucial test case for ongoing debates about the centrality of predictive processing in language comprehension, especially for algorithmic-implementational accounts.[3,7–9] However, previous studies are limited in that they have not jointly manipulated contextual predictability and the pragmatics of negation. This is crucial, since the pragmatic requirements of negation have been argued to influence not only the cost of comprehending negation but also the algorithm used to do so.[10] Thus, fully describing prediction's role in negation processing requires an experimental paradigm that addresses both factors simultaneously. We developed just such a paradigm here.

In a sentence verification task, English-speaking participants (N = 75) viewed displays containing two geometric shapes that varied in color only (e.g., blue vs. red) or in color and texture (e.g., blue solid vs. red dotted). Participants were told that these displays were being used to train a robot to identify shapes. The robot printed one of the shapes, and a fictional human supervisor provided feedback in the form of affirmative or negative sentences. The supervisor was unreliable, and the participant's job was to quickly evaluate the feedback as true or false (Figure 1). Crucially, the robot's errors created the opportunity for *corrections*—a felicitous context for the use of negative sentences[11] (in contrast to infelicitous contexts, where the robot reproduced the shape correctly). The number of visual features that varied between the two shapes (1 vs. 2) modulated the predictability of the supervisor's statements. Pragmatic context and predictability, in this sense, were operationalized as binary variables (felicitous vs. infelicitous and high vs. low, respectively). These variables were fully crossed with the polarity of the supervisor's feedback (affirmative vs. negative) and its truth value (true vs. false). Correct response times (RTs) from the onset of the supervisor sentence were our dependent variable. We hypothesized that, if predictability or pragmatic context influenced negation comprehension, they would modulate the crossover interaction between sentence polarity and truth value.

This prediction was borne out for pragmatic context (three-way interaction with polarity and truth value: $\chi^2(1) = 38.9$, $p < .0001$, Figure 2, top vs. bottom panels). In felicitous contexts, the crossover nature of the interaction entirely disappeared: true and false negatives did not differ (Figure 2, top, red; both p's > .99), while true affirmatives were evaluated significantly faster than false ones (all p's < .0001). In infelicitous contexts, the classic crossover interaction was preserved (all p's < .0001). These patterns all held regardless of predictability level. Predictability, by contrast, showed no significant interactions ($\chi^2(1) = .54$, $p = .46$) but showed a robust main effect ($\beta = -.04$, SE = .005, t = -7.00, $p < .0001$). This demonstrates a general facilitation for high-predictability trials independent of sentence structure, including negation.

Taken together, our results are in line with a large body of literature that highlights the decisive role of pragmatic context in negation processing. Contextual predictability, however, yielded only a global speed-up, suggesting that a more general predictive mechanism may be at play, rather than one that is proactively involved in the integration of logical operators into sentence meaning.

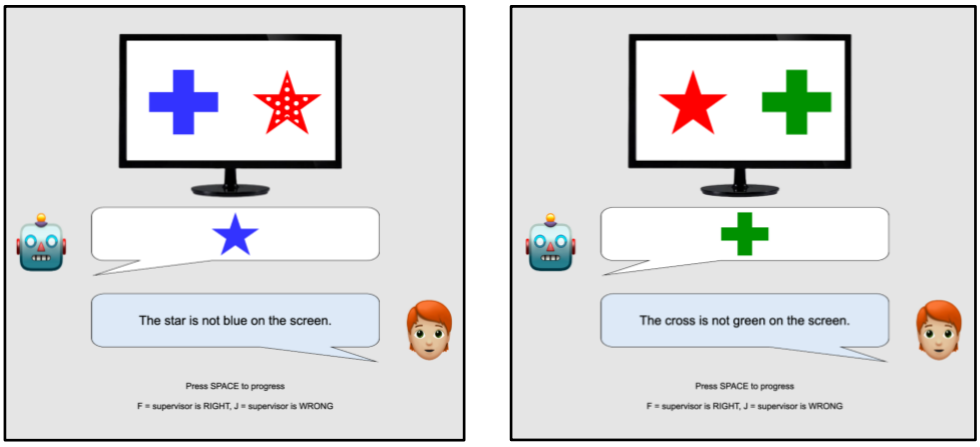

Figure 1: Example trial screens for felicitous (left) and infelicitous (right) trials. Felicity refers to whether a correction by the supervisor is warranted or not, based on Shapo's output. In low-predictability trials (left), the geometric shapes varied in both color and texture; in high-predictability trials (right), they varied in color only. For low-predictability trials, color and texture were mentioned by the supervisor an equal number of times. The other manipulated variables were the polarity (affirmative vs. negative) and truth value (true vs. false) of the supervisor's statements.

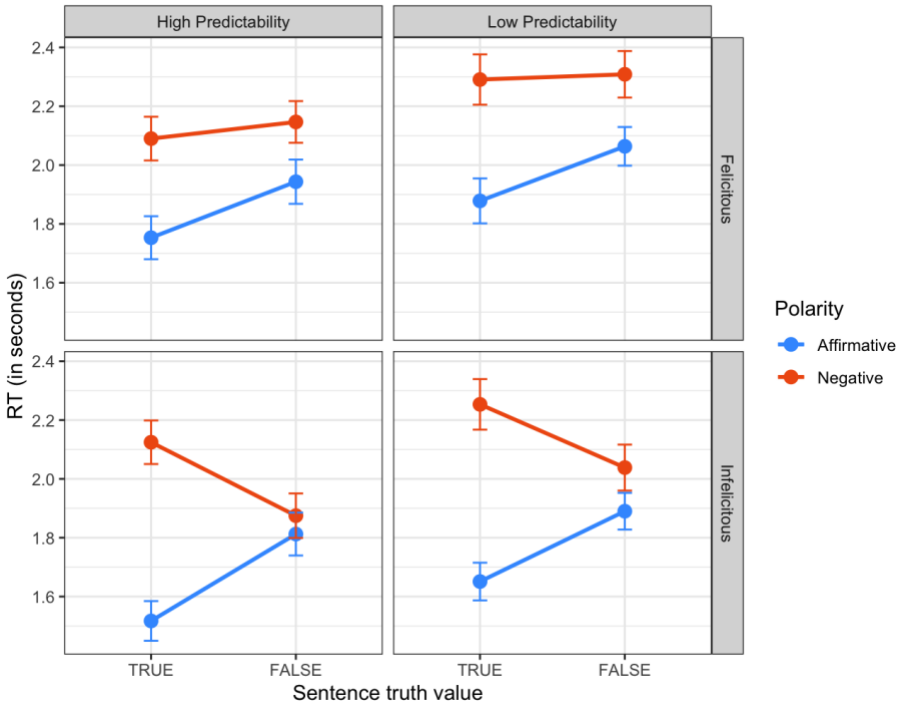

Figure 2: Mean RT per condition, error bars represent 95% within-subject CIs. 'Felicitous' and 'Infelicitous' refer to the levels of the pragmatic context type variable. 'High' and 'low predictability' refer to the levels of predictability of the critical word in the supervisor's sentence (e.g., *The star is/is not blue on the screen.*).

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